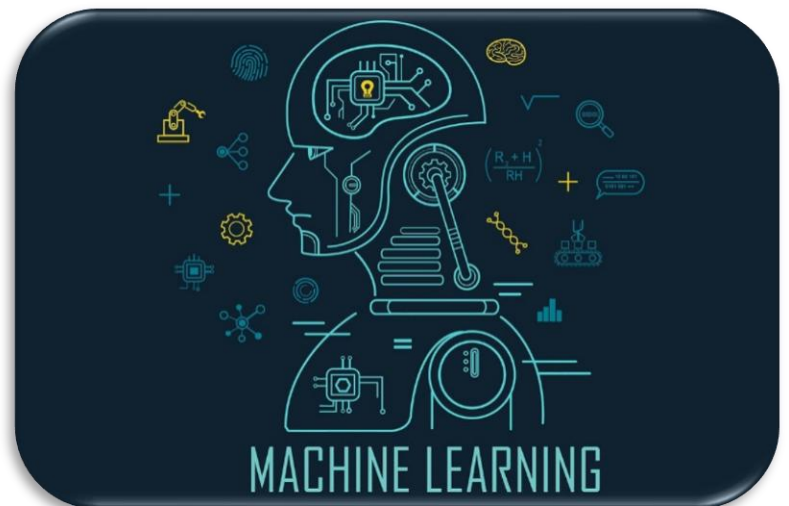


Introduction to Machine Learning



MODULE 2: MACHINE LEARNING

In today's digital era, computers are used not only to perform simple calculations but also to solve complex problems such as recognizing images, understanding speech, and making predictions. With the rapid growth of data, it has become essential for systems to learn from data and improve their performance automatically. Machine Learning plays a key role in developing such intelligent systems. This SLM introduces the fundamental ideas of Machine Learning and helps learners understand how it is applied in real-world situations.

2.1 What is Machine Learning?

Machine learning is a branch of artificial intelligence where computers learn from data and improve their performance without being told exactly what to do every time. In simple words, **machine learning teaches computers to learn from experience**, just like humans do. For example: If you show a computer thousands of pictures of cats and dogs, it will learn the differences between them. Later, when you show it a new picture, it can tell whether it's a cat or a dog.



Pause and Reflect



Why do you think computers need to learn from data instead of being programmed with fixed rules for every task?



Think & Write

How is learning from pictures similar to how humans learn to recognize objects?

2.1.1 Key Ideas of Machine Learning

- **Learning from data:** Algorithms analyze large datasets to detect patterns humans might miss. It studies a large amounts of data to find patterns which will later guide him while taking decisions for you or making predictions.

Example:

- ✓ YouTube looks at videos you watch.
- ✓ It learns what you like.
- ✓ Then it recommends similar videos.

- **Prediction and decision-making:** Once trained, the system can make predictions.

Example:

- ✓ Predicting tomorrow's weather
- ✓ Predicting if a student might need extra help
- ✓ Detecting if a bank transaction is fraud

- **Adaptability:** Systems evolve as new data arrives, staying relevant in changing environments. More data = Better learning.

Example:

- ✓ Google Maps learns traffic patterns.
- ✓ It gives better route suggestions over time.

- **Applications:** Machine learning is Used in image recognition, speech processing, recommendation systems, fraud detection, autonomous vehicles, and more.



Pause & Reflect

Which key idea of Machine Learning do you think is most important in daily life and why?

Before you learn more about ML, you definitely should know what do we mean by “data”.. If you don’t understand data, ML will feel magical and confusing.

2.1.2 What is Data?

Data is simply information. It can be numbers, words, pictures, sounds, or any facts that we collect to learn something. In Machine learning: Data is the raw information that a computer uses to learn.



Easy Everyday Examples :

Student Marks, If you write down marks of students:

Name	Marks(out of 50)
Kunal	20
Deep	30

These marks are data and the computer can use this data to:

- ✓ Find the average
- ✓ Predict future performance
- ✓ Identify who needs help

Simple Definition for Students would be : Data is a collection of facts, numbers, pictures, or information that a computer uses to learn and make decisions.



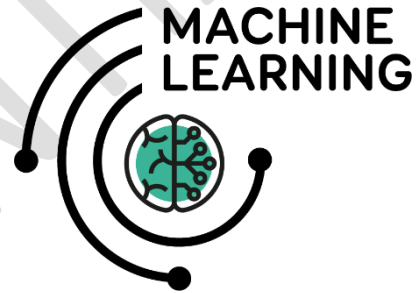
Pause and Reflect

If wrong or incomplete data is given to a computer, how might it affect the results?

2.2 Types of Machine Learning

Machine Learning is mainly divided into **three main types**:

- i. Supervised Learning
- ii. Unsupervised Learning
- iii. Reinforcement Learning



Type	How it works	Example
Supervised learning	Learns from labelled data	Predicting house prices
Unsupervised learning	Finds hidden patterns in unlabeled data	Customer segmentation
Reinforcement learning	Learns by trial and error with rewards	Training robots or game AI

a. **Supervised Learning** : The computer learns from labeled data.

Here Labeled data means:

- We give the input, Example - We give Pictures of Dog and cats.
- And we also give the correct answer, example - We mention which of them are dogs and which are cats.
- It's like learning with a teacher who provides both questions and answers.

In Supervised learning : The computer learns using examples that already have correct answers.

Used for:

- ✓ Predicting house prices

- ✓ Spam email detection
- ✓ Medical diagnosis



Think & Write

Give one example from your daily life where supervised learning could be used.

b. **Unsupervised Learning** : The computer learns from unlabeled data.

Here Unlabeled data means :

- We only give the data, Example : data of how different customer profile who comes for shopping in a grocery mart.
- No correct answers are provided, Example : No label provided as on - what type of customer etc.
- The computer finds patterns or groups on its own. Example : It analyzes customers and groups them into: Frequent buyers or Premium customers

In Unsupervised learning : No one tells the computer how to group them, it figures it out.

Used for:

- ✓ Customer segmentation
- ✓ Grouping similar products
- ✓ Reducing large data into smaller meaningful parts



Pause & Reflect

Why do you think unsupervised learning is useful when data labels are not available?



Quick Check

Match the following:

- Learning with labels → _____
- Learning without labels → _____
- Learning with rewards → _____

2.2.1 Supervised vs Unsupervised Learning.

Let's understand the difference in a more elaborate way...

Aspect	Supervised learning	Unsupervised learning
Data type	Uses labeled data (input + correct output)	Uses unlabeled data (only input, no output)

Goal	Learn the mapping between inputs and outputs	Discover hidden patterns or groupings
Examples	Predicting house prices, spam detection, medical diagnosis	Customer segmentation, anomaly detection, market basket analysis
Approach	Teacher-like guidance: algorithm knows the “right answers” during training	Exploratory: algorithm finds structure without guidance
Algorithms	Linear regression, decision trees, support vector machines, neural networks	K-means clustering, hierarchical clustering, PCA (dimensionality reduction)
Output	Predictions or classifications	Clusters, associations, or reduced dimensions
Use cases	When labeled data is available and accuracy matters	When data is abundant, labels are scarce or costly

Quick Analogy

- **Supervised learning** = A student learning with an answer key.
- **Unsupervised learning** = A student exploring patterns without answers, grouping things by similarity.



Quick Check

1. Which type of learning uses labeled data? _____
2. Which type focuses on finding hidden patterns? _____

c. **Reinforcement Learning** : The computer learns by trial and error.

To simply understand it - The system performs actions and: *Gets a reward if correct & Gets a penalty if wrong*. And by repeating it with multiple case scenarios or different data sets with time, it learns the best decisions. Meaning - It learns from it's mistake over the time.

Example:

- i. Game-playing AI(chess bot etc.)
- ii. Self-driving cars

In short Reinforcement learning is Learning by practice and rewards.

Now that we understand how machines learn, let us see where this learning is applied and how the data for learning is collected and prepared.



Think and Relate

How is reinforcement learning similar to how humans learn from rewards and mistakes?

2.3 Machine Learning Workflow

The **Machine Learning workflow** describes the step-by-step process followed to build an intelligent system. Machine learning does not work alone. Before a computer can learn, we must:

- i. **Collect data**
- ii. **Understand and organize it**
- iii. **Train a model**
- iv. **Apply it to real-world problems like language and images**

Let's understand each step.

i. **Data Acquisition(Collecting Data)**

Data Acquisition is the process of collecting and measuring information from real-world sources so it can be analyzed, stored, and used in applications like machine learning, computer vision, or scientific research. It's essentially the "first step" in turning raw signals or observations into usable digital data.

In Simple term - Data Acquisition means collecting information from the real world so the computer can use it.

Example - Collecting books before studying for a particular exam.

➤ **Key Elements of Data Acquisition**

- **Sensors and instruments:** Devices that capture signals (e.g., temperature sensors, cameras, microphones).
- **Sampling:** Converting continuous signals into discrete data points at regular intervals.
- **Digitization:** Transforming analog signals into digital form for processing.
- **Storage:** Saving acquired data in databases, cloud systems, or local devices.
- **Preprocessing:** Cleaning, filtering, or normalizing data before analysis.

➤ **Types of Data Acquisition**

Sr. No.	Type	Description	Example
1	Manual acquisition	Human-driven collection	Surveys, manual measurements
2	Automated acquisition	Machine-driven, continuous collection	IoT sensors, smart meters
3	Batch acquisition	Data collected in chunks	Daily transaction logs
4	Real-time acquisition	Instantaneous data capture	Stock market feeds, live video

➤ **Tools & Technologies**

- **DAQ systems:** Hardware/software platforms for capturing sensor data.
- **IoT devices:** Smart sensors connected to networks.
- **APIs & web scraping:** Collecting digital data from online sources.
- **Cloud platforms:** Storing and streaming acquired data for analysis.

➤ Applications

- **Healthcare:** Patient monitoring via wearable sensors.
- **Manufacturing:** Machine performance tracking.
- **Finance:** Real-time market data feeds.
- **Research:** Scientific experiments and environmental monitoring.



Pause and Reflect

Why do you think data collection is the most important step in the ML workflow?

ii. Data Exploration (Understand and organize it)

Data Exploration is the stage in data analysis where you examine datasets to understand their structure, spot patterns, detect anomalies, and generate initial insights before applying advanced models. It's like "getting to know your data" before deciding how to use it. Meaning, we must check:

- ✓ Is it correct?
- ✓ Is anything missing?
- ✓ What patterns are present?

Example: Before predicting exam results, we check: *Average marks, Highest and lowest scores and Missing values*

This helps us understand the data better.

➤ Key Goals of Data Exploration

- **Understand structure:** Identify variables, data types, and relationships.
- **Detect quality issues:** Spot missing values, duplicates, or outliers.
- **Summarize data:** Use descriptive statistics (mean, median, variance).
- **Visualize patterns:** Charts and plots reveal trends and distributions.
- **Guide next steps:** Helps decide which models or preprocessing techniques to apply.

➤ Tools Commonly Used

- **Python libraries:** Pandas, NumPy, Matplotlib, Seaborn.
- **R packages:** ggplot2, dplyr.
- **BI tools:** Power BI, Tableau.
- **SQL queries:** For structured database exploration.

➤ Real-World Applications

- **Business analytics:** Understanding customer purchase behavior.
- **Healthcare:** Exploring patient records for risk factors.
- **Finance:** Detecting unusual transactions.
- **Science:** Analyzing experimental data before modeling.



Pause & Reflect

What might happen if we skip data exploration and directly train a model?

iii. Data Modelling (Train a model)

Data Modeling is the process of creating a structured representation of data, showing how different pieces of information relate to each other. It's essentially the blueprint for organizing, storing, and using data effectively in databases, analytics, and machine learning. In simple terms, we organize the data properly by creating a structured plan for storing and connecting data.

Example: In a school system: we create multiple table to store different types of data but they should have some common field that links them together.

- ✓ Student table
- ✓ Subject table
- ✓ Marks table

➤ Key Goals of Data Modeling

- **Organization:** Define how data is stored and connected.
- **Clarity:** Make complex data easier to understand.
- **Consistency:** Ensure data follows rules and standards.
- **Efficiency:** Optimize storage and retrieval for performance.

➤ Types of Data Models

Sr. No.	Type	Description	Example
1	Conceptual model	High-level overview of data and relationships	ER diagrams showing customers and orders
2	Logical model	More detailed, defines attributes and relationships	Tables with fields like CustomerID, OrderDate
3	Physical model	Actual implementation in a database	SQL schema with indexes and constraints

➤ Real-World Applications

- **Business systems:** Customer relationship management (CRM).
- **Finance:** Transaction records and audit trails.
- **Healthcare:** Patient records and medical histories.

- **AI/ML:** Structuring training datasets for predictive models.



Think & Write

Why is organizing data important before training a machine learning model?

4.LP and Computer Vision(Apply it to real-world problems like language and images). Once data is collected and organized, we train the computer using: **a) Supervised Learning, b) Unsupervised Learning and c) Reinforcement Learning.**

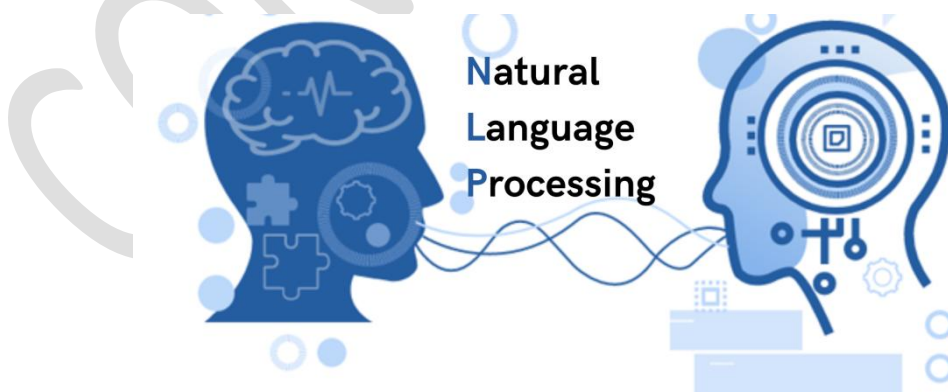
After training, the computer can now make predictions or decisions. To make machines interact more naturally with humans and understand the world better.

2.4 Applications of Machine Learning

Machine Learning is widely used in many real-world domains to solve complex problems and automate tasks. Some common applications include: **Natural Language Processing (NLP)** and **Computer Vision (CV)** Let's understand each in detail...

A. Natural Language Processing

Natural Language Processing (NLP) is a field of artificial intelligence that enables computers to understand, interpret, and generate human language in text or speech. It bridges the gap between human communication and computer systems, making machines capable of interacting with us in natural ways. Natural Language Processing (NLP) helps computers understand and use human language (text or speech). In simple words: NLP teaches computers to read, understand, and speak human language.



➤ Core Concepts of NLP

- **Syntax analysis:** Understanding grammar and sentence structure.
- **Semantic analysis:** Extracting meaning from words and phrases.

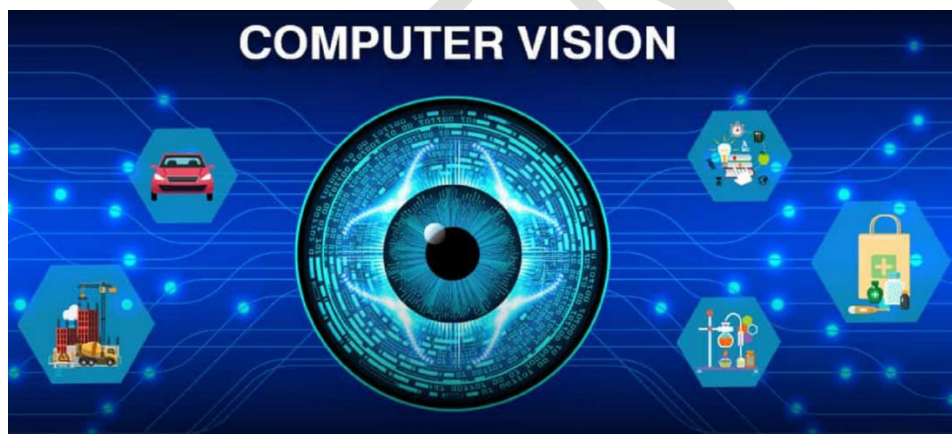
- **Pragmatics:** Interpreting context and implied meaning.
- **Speech recognition:** Converting spoken language into text.
- **Natural language generation:** Producing human-like text or speech.

➤ Real-World Applications

- **Chatbots and virtual assistants:** Siri, Alexa, Copilot.
- **Machine translation:** Google Translate, DeepL.
- **Sentiment analysis:** Understanding emotions in social media posts.
- **Text summarization:** Condensing long articles into key points.
- **Information retrieval:** Search engines finding relevant results.

B. Computer Vision(CV)

Just like NLP helps computers understand language, **Computer Vision helps them understand images.** Computer Vision (CV) is a field of artificial intelligence that enables machines to interpret and understand visual information from the world, such as images and videos. It's about teaching computers to "see" and make sense of what they see, much like humans do. In simple words : Computer Vision teaches computers to understand images like humans do.



➤ Core Concepts in Computer Vision

- **Image classification:** Identifying what an image contains (e.g., cat vs. dog).
- **Object detection:** Locating and labeling multiple objects within an image.
- **Image segmentation:** Dividing an image into meaningful regions (e.g., separating background from foreground).
- **Feature extraction:** Detecting edges, shapes, or textures to analyze visual data.
- **Face recognition:** Identifying or verifying individuals from facial features.

➤ Real-World Applications

- **Healthcare:** Detecting tumors in medical scans.

- **Autonomous vehicles:** Recognizing pedestrians, traffic signs, and obstacles.
- **Retail:** Automated checkout systems and inventory tracking.
- **Security:** Surveillance and facial recognition.
- **Agriculture:** Monitoring crop health using drone imagery.

Natural Language Processing helps computers understand and respond to human language, while Computer Vision enables computers to interpret images and videos.



Pause & Reflect

Which application of Machine Learning do you think will be most useful in the future and why?

CONFIDENTIAL